Google Ads API Access Design Document

Application Summary

Cap is applying for Google Ads API Basic Access for an internal advertising operations tool used only to manage Cap's own Google Ads account.

The tool is a locally hosted Model Context Protocol (MCP) server based on Google's open source `google-ads-mcp` project, extended with guarded tools for campaign planning, reporting, recommendations, conversion upload, and campaign/ad management. It is not a third-party SaaS product and is not offered to external advertisers.

Requested access level: Basic Access.

Requested permissible use:

- Reporting
- Researching keywords and recommendations
- Ad creation / management

Primary Google Ads customer ID: `376-913-4291`

Business Use Case

Cap currently has organic revenue and wants to test paid acquisition through Google Ads. The API will be used to:

- Research search demand, keyword volume, competition, and estimated CPCs.
- Forecast proposed keyword plans before launching campaigns.
- Create and maintain small, tightly scoped Search campaigns for Cap.
- Review search-term performance and add negative keywords.
- Monitor campaign, budget, ad, and conversion performance.
- Upload server-side/offline purchase conversions when Google click IDs are available.
- Keep a human-reviewed audit trail of suggested and applied changes.

The tool is intended to reduce manual campaign setup and reporting work while keeping final launch, enablement, and budget decisions under human review.

Users and Access Model

The only users are authorized Cap operators with access to Cap's Google Ads account and local development environment. The MCP server runs locally on a trusted machine and authenticates through OAuth Application Default Credentials for a Google account with access to the target Google Ads account.

The system does not create Google Ads accounts, manage user invitations, or modify billing settings.

OAuth scope:

- `https://www.googleapis.com/auth/adwords`

System Architecture

The integration has four components:

1. Codex or another MCP-capable agent acts as the operator interface.
2. A local MCP server exposes typed Google Ads tools.
3. The MCP server uses the official Google Ads Python client library.
4. Google Ads API executes read-only requests, validation-only mutation requests, or approved mutation requests.

The MCP server is configured locally with:

- OAuth client JSON for browser-based login.
- Application Default Credentials generated by `gcloud auth application-default login`.
- Google Ads developer token.
- Optional manager login customer ID.
- Local guardrail environment variables for mutation access.

API Services and Methods

Account Discovery and Metadata

Purpose: Identify accessible accounts and discover field compatibility.

Services:

- `CustomerService.ListAccessibleCustomers`
- `GoogleAdsFieldService.SearchGoogleAdsFields`

MCP tools:

- `customers_list_accessible_customers`
- `metadata_get_resource_metadata`

Reporting

Purpose: Pull performance data and operational state for Cap's own account.

Services:

- `GoogleAdsService.SearchStream`

Resources queried:

- `customer`
- `campaign`
- `campaign_budget`
- `ad_group`
- `ad_group_ad`
- `search_term_view`
- `change_event`

MCP tools:

- `search_search`
- `reports_search_terms_report`
- `reports_negative_keyword_recommendations`
- `reports_change_history`
- `reports_ad_policy_and_strength_report`
- `reports_budget_pacing_report`

Example reporting fields:

- Campaign and ad group names and IDs.
- Impressions, clicks, cost, conversions, and conversion value.
- Search terms.
- Ad policy summaries, primary statuses, ad strength, and action items.
- Change event resource type, operation, user email, and changed fields.

Keyword Planning and Forecasting

Purpose: Evaluate candidate keywords before spend is committed.

Services:

- `KeywordPlanIdeaService.GenerateKeywordIdeas`
- `KeywordPlanIdeaService.GenerateKeywordHistoricalMetrics`
- `KeywordPlanIdeaService.GenerateKeywordForecastMetrics`

MCP tools:

- `planning_generate_keyword_ideas`
- `planning_get_keyword_historical_metrics`
- `planning_forecast_keyword_plan`

Inputs:

- Seed keywords and/or Cap landing page URLs.
- Geo target constants.
- Language constants.
- Match type.

- Budget and bid assumptions.

Outputs:

- Suggested keywords.
- Historical monthly search volume.
- Competition and bid estimates.
- Forecasted impressions, clicks, cost, conversions, and conversion value when available.

Recommendations

Purpose: Retrieve Google Ads recommendations for advisory review only.

Services:

- `RecommendationService.GenerateRecommendations`

MCP tools:

- `recommendations_generate_recommendations`

Google-generated recommendations are never auto-applied by the current tool. They are treated as inputs to a human-reviewed change plan.

Campaign and Ad Management

Purpose: Create and maintain Search campaigns, ad groups, keywords, ads, budgets, and negative keywords for Cap's own account.

Services:

- `CampaignBudgetService.MutateCampaignBudgets`
- `CampaignService.MutateCampaigns`
- `AdGroupService.MutateAdGroups`
- `AdGroupCriterionService.MutateAdGroupCriteria`
- `CampaignCriterionService.MutateCampaignCriteria`
- `AdGroupAdService.MutateAdGroupAds`

MCP tools:

- `mutations_create_campaign_budget`
- `mutations_create_paused_search_campaign`
- `mutations_create_ad_group`
- `mutations_add_ad_group_keywords`
- `mutations_create_responsive_search_ad`
- `mutations_add_campaign_negative_keywords`
- `mutations_set_campaign_status`
- `mutations_set_ad_group_status`
- `mutations_set_ad_group_ad_status`

- `mutations_set_campaign_budget_amount`
- `mutations_update_responsive_search_ad`
- `mutations_get_mutation_guardrails`

Design constraints:

- Campaign creation creates paused Search campaigns by default.
- Ad groups, keywords, and ads default to paused where applicable.
- Mutations default to `validate_only=true`.
- Real mutation execution requires an explicit local environment flag.
- Real `ENABLED` status changes require a second explicit launch flag.
- Optional local budget and CPC caps block oversized changes before any API call.

Offline Conversion Upload

Purpose: Send Cap purchase or subscription conversion events back to Google Ads when click IDs are available.

Services:

- `ConversionUploadService.UploadClickConversions`

MCP tools:

- `conversions_upload_click_conversions`

Inputs:

- Exactly one of `gclid`, `gbraid`, or `wbraid`.
- Conversion action.
- Conversion date/time.
- Conversion value.
- Currency code.
- Optional order ID for deduplication.

Design constraints:

- Uploads default to `validate_only=true`.
- Real uploads require the same explicit mutation environment flag as other writes.
- Uploads are intended only for Cap's own conversion events.

Human Review and Safety Controls

The tool is designed to be human-in-the-loop.

Default behavior:

- Read-only tools are enabled.
- Write-capable namespaces are visible for dry-run validation.

- All write tools default to `validate_only=true`.
- Real writes are blocked unless `GOOGLE_ADS_MCP_ENABLE_MUTATIONS=true` is set.
- Enabling campaigns, ad groups, keywords, or ads is blocked unless `GOOGLE_ADS_MCP_ALLOW_ENABLE=true` is set.

Optional guardrail variables:

- `GOOGLE_ADS_MCP_MAX_DAILY_BUDGET_MICROS`
- `GOOGLE_ADS_MCP_MAX_CPC_BID_MICROS`

Operational workflow:

1. Pull reporting and planning data.
2. Generate a proposed plan.
3. Run mutation tools in `validate_only=true`.
4. Review expected changes, affected customer ID, budget, bids, and campaign/ad statuses.
5. Apply only explicitly approved changes.
6. Monitor change history and performance after launch.

Data Handling

The tool processes Google Ads account data only for Cap's own account. Data is used for campaign planning, reporting, optimization, and conversion attribution.

The tool does not sell Google Ads data, expose it to external customers, or use it to create audience profiles outside Cap's advertising operations.

Local storage:

- OAuth credentials are stored by Google Cloud SDK as Application Default Credentials.
- Developer token and customer IDs are stored in a local environment file with restricted file permissions.
- The tool does not maintain an application database of Google Ads data.
- Any reporting output is transient in the operator session unless manually saved by an operator.

Secrets:

- OAuth client secret and developer token are not committed to source control.
- Local env files are outside the repository.
- File permissions are restricted to the local user.

Out-of-Scope API Use

The tool does not use the Google Ads API for:

- Creating Google Ads customer accounts.
- Managing account users or invitations.

- Changing billing or payments settings.
- Accessing invoices or payment accounts.
- Managing third-party advertiser accounts.
- Automatically applying Google recommendations.
- Launching or increasing spend without human approval.

Quota and Rate-Limit Behavior

The expected initial usage is low-volume and internal:

- A small number of keyword planning and reporting calls during campaign setup.
- Daily or ad hoc reporting checks.
- Occasional campaign, ad, budget, keyword, and negative keyword changes.
- Occasional conversion uploads if eligible click IDs exist.

Basic Access should be sufficient for the initial workflow. If Cap later needs a higher operation limit, it will apply for Standard Access separately.

The application will handle Google Ads API errors by surfacing them to the operator. Rate-limit and quota errors will stop the current action and require the operator to retry later or reduce scope.

Required Minimum Functionality

This application is requesting Basic Access for internal use. It is not currently applying for Standard Access and is not offered as an external tool to advertisers. If Cap later applies for Standard Access or externalizes the tool, Cap will review and implement any Required Minimum Functionality obligations before applying.

Reviewer Notes

This is an internal operations tool for a single advertiser account. The requested API access is needed because the current developer token is limited to test accounts and cannot validate or manage Cap's production Google Ads account.

The implementation intentionally separates read-only analysis from write-capable operations and uses local environment flags, validation-only requests, paused defaults, and human review to reduce the risk of accidental campaign launch or spend changes.

References

- Google Ads API access levels and permissible use: https://developers.google.com/google-ads/api/docs/api-policy/access-levels
- Google Ads API access levels and production guidance: https://developers.google.com/google-ads/api/docs/productionize/access-levels

- Google Ads API OAuth single-user authentication: https://developers.google.com/google-ads/api/docs/oauth/single-user-authentication
- Google Ads API developer token: https://developers.google.com/google-ads/api/docs/get-started/dev-token